

Course: B. Tech.

Branch: IT

Semester: III

Subject Code & Name: Data Structure and Application (BTITC305)

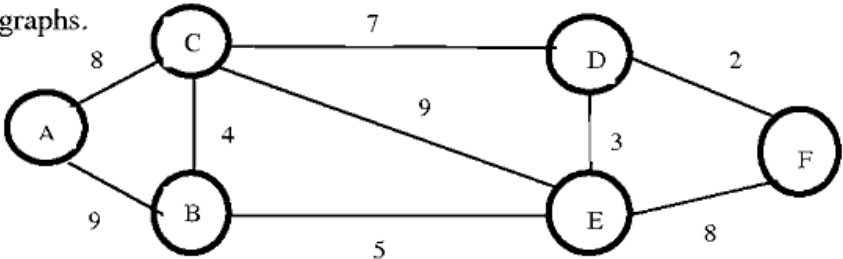
Max Marks: 60

Date: 21/08/2023

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Describe Time and Space complexity related to an algorithm.	CO1	06
B) Write the definition of data structures. What are the different types of data structures?	CO1	06
C) With an example explain Asymptotic Notation along with it types.	CO1	06
Q.2 Solve Any Two of the following.		12
A) Explain representation of linked lists in the memory.	CO2	06
B) Draw and explain the circular queue with an example in detail.	CO2	06
C) Develop an algorithm to insert a node at the end of the linked list.	CO3	06
Q. 3 Solve Any Two of the following.		12
A) Explain Stack data structure. Write an algorithm of PUSH and POP operations of stack.	CO4	06
B) What is a binary tree? Write any three types of binary tree with an example.	CO3	06
C) Explain in detail binary tree traversal techniques.	CO3	06
Q.4 Solve Any Two of the following.		12
A) Explain Prim's algorithm with an example?	CO4	06
B) Find the minimum spanning tree using Kruskal's algorithm for the given graphs.	CO2	06
		
C) Explain Breadth First Search (BFS) technique along with an algorithm.	CO4	06

Q. 5	Solve Any Two of the following.		12
A)	How does the Quick sort works? Explain with the help of example.	CO5	06
B)	Sort the following list using Merge Sort. Write its complexity. 12, 8, 2, 6, 25, 5, 4, 9.	CO5	06
C)	Write an algorithm for linear search in an array.	CO2	06

***** End *****